Supermarket Sales Analysis

Data Analytics Project

1. Problem Statement

The aim of this project is to analyze supermarket sales data and find useful information about products, branches, categories, customers, payments, and ratings.

2. Dataset

Dataset

The dataset contains 500 sales transactions. It includes product, branch, city, customer type, quantity, price, payment method, rating, and sales.

3. Steps Used for Analysis

1. Collect and load the CSV dataset.
2. Check the data for missing or incorrect values.
3. Calculate Sales = Quantity × Unit Price.
4. Group and summarize the data using totals, counts, and averages.
5. Create charts to compare the results.
6. Use the results to make business decisions.

4. Live Analysis

Which product generates the highest sales?

(Cheese generated the highest sales: ₹27,906.30.)

Which branch performs best?

Branch C (Mumbai) performed best with sales of ₹72,469.45.

Which category sells the most?

Beverages had the highest sales with ₹56,108.24.

What is the most popular payment method?

UPI was the most used payment method with 127 transactions.

Do Members spend more than Normal customers?

No. The average Member transaction was ₹483.14, while Normal customers averaged ₹497.07.

What is the average customer rating?

The average customer rating was 3.99 out of 5.

5. Business Decisions

The supermarket can keep more stock of high-selling products and categories, study the reasons for Branch C's strong performance, continue supporting UPI payments, and improve customer service to increase ratings. Customer spending can also be used to plan membership offers.

6. Conclusion

This analysis shows how sales data can be converted into simple insights that help a supermarket understand performance and make better decisions.